

ISMCE

(In-System Memory Content Editor)

Quartus configuration to VHDL 2008 version

Settings - Test

Category:

- General
- Files
- Libraries
- ▼ IP Settings
 - IP Catalog Search Locations
- Design Templates
- ▼ Operating Settings and Conditions
 - Voltage
 - Temperature
- ▼ Compilation Process Settings
 - Incremental Compilation
- ▼ EDA Tool Settings
 - Design Entry/Synthesis
 - Simulation
 - Board-Level
- ▼ **Compiler Settings**
 - VHDL Input**
 - Verilog HDL Input
 - Default Parameters
- Timing Analyzer
- Assembler
- Design Assistant
- Signal Tap Logic Analyzer
- Logic Analyzer Interface
- Power Analyzer Settings
- SSN Analyzer

VHDL Input

Options for directly compiling or simulating VHDL input files. (Click on the EDA Tool

VHDL version

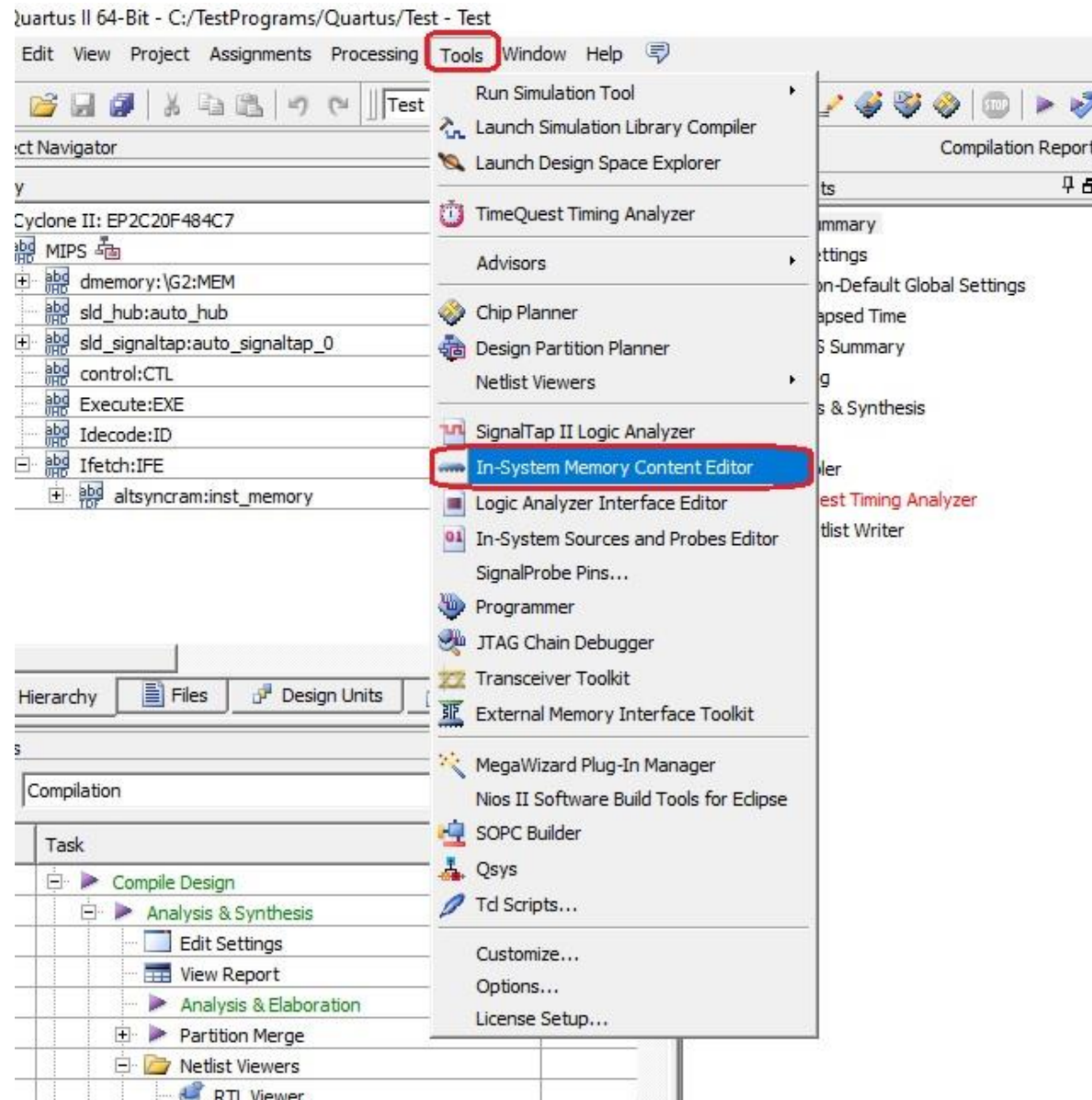
- ☐ VHDL 1987
- ☐ VHDL 1993
- ☒ **VHDL 2008**

Library Mapping File

File name:

☐ Show information messages describing LMF mapping during compilation

Open ISMCE



Digital System Design Loading

In-System Memory Content Editor - C:/TestPrograms/Quartus21_1/Test - Test

File Edit View Processing Tools Window Help

Instance Manager:

No instances found

Index	Instance ID	Status	Width	Depth	Type	Mode
-------	-------------	--------	-------	-------	------	------

No instance...

Hardware Setup

Hardware Settings

JTAG Settings

Select a programming hardware setup to use when programming devices. This programming hardware setup applies only to the current programmer window.

Currently selected hardware: No Hardware

Hardware frequency: DE-SoC [USB-1]

Available hardware items

Hardware	Server	Port
DE-SoC	Local	USB-1

Add Hardware...

Remove Hardware

Close

Search Intel FPGA

JTAG Chain Configuration: JTAG ready

Hardware: DE-SoC [USB-1]

Setup...

Device: @1: SOCVHPS (0x4BA)

Scan Chain

File: C:/TestPrograms/Quartus21_1/output_files/Test.sof

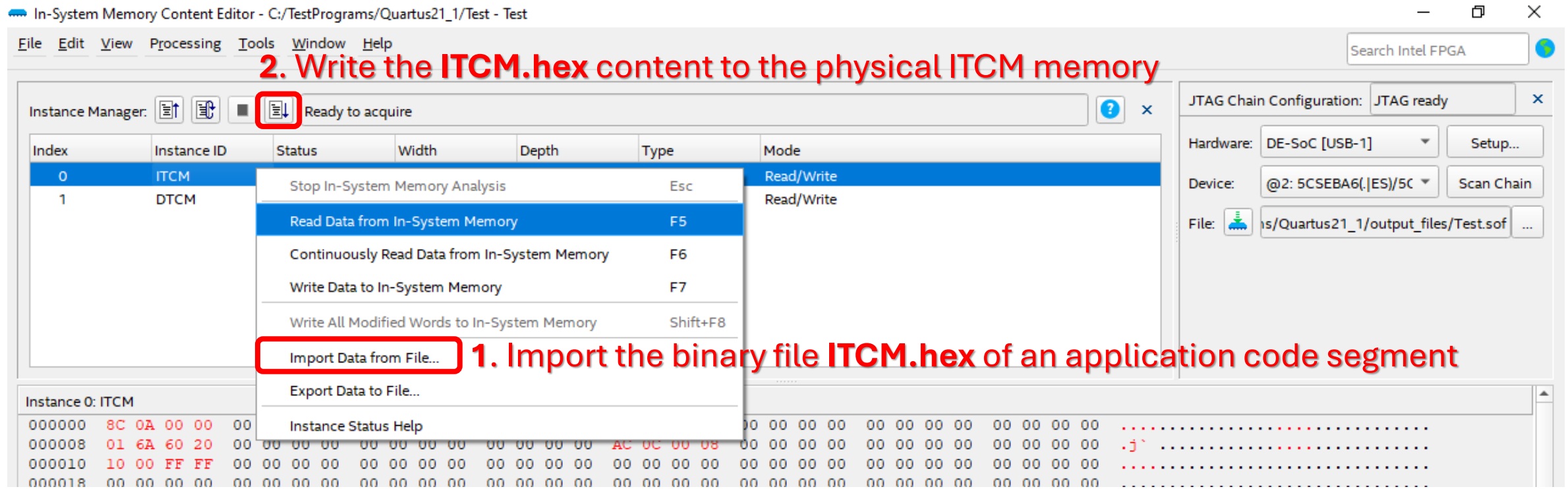
2. Load the design file *.sof to the FPGA chip only once per design cycle

1. Set the HW connection

Validation using ISMCE

Running applications (recurring steps)

Step 1 – Load the application's code segment bin file



Step 2 – Load the application's data segment bin file

In-System Memory Content Editor - C:/TestPrograms/Quartus21_1/Test - Test

File Edit View Processing Tools Window Help

Search Intel FPGA

2. Write the DTCM.hex content to the physical DTCM memory

Instance Manager: Ready to acquire

Index	Instance ID	Status	Width	Depth	Type	Mode
0	ITCM	Not running	32	256	RAM/ROM	Read/Write
1	DTCM	Not running	32	256	RAM/ROM	Read/Write

- Stop In-System Memory Analysis Esc
- Read Data from In-System Memory F5
- Continuously Read Data from In-System Memory F6
- Write Data to In-System Memory F7
- Write All Modified Words to In-System Memory Shift+F8
- Import Data from File...
- Export Data to File...
- Instance Status Help

1. Import the binary file DTCM.hex of an application code segment

Instance 0: ITCM

000000	8C 09 00 00 8C 0A	0D 00 10 01 6C 68 20 AC 0D 00 14 10 00 FF FF	*hlh
00000d	00 00 00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00 00	
00001a	00 00 00 00 00 00	00 00 00 00 00 00 00 00 00 00 00 00 00	

JTAG Chain Configuration: JTAG ready

Hardware: USB-Blaster [USB-0] Setup...

Device: @1: 10CL120(Y|Z)/EP: Scan Chain

File: s/Quartus21_1/output_files/Test.sof ...

1. Click the RESET pushbutton (KEY0) to run the application

1. Click the RESET pushbutton (KEY0) to run the application

2. Choose the **DTCM** memory

Step 4 – Validation using the golden model

In-System Memory Content Editor - C:/TestPrograms/Quartus21_1/Test - Test

File Edit View Processing Tools Window Help

Search Intel FPGA

Instance Manager: Ready to acquire

Index	Instance ID	Status	Width	Depth	Type	Mode
0	ITCM	Not running	32	256	RAM/ROM	Read/Write
1	DTCM	Not running	32	256	RAM/ROM	Read/Write

Stop In-System Memory Analysis Esc

Read Data from In-System Memory F5

Continuously Read Data from In-System Memory F6

Write Data to In-System Memory F7

Write All Modified Words to In-System Memory Shift+F8

Import Data from File

Export Data to File...

Instance Status Help

Instance 0: ITCM

000000 8C 09 00 00 8C 02 0D 00 10 01 6C 68 20 AC 0D 00 14 10 00 FF FF |*hlh
00000d 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |
00001a 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |

JTAG Chain Configuration: JTAG ready

Hardware: USB-Blaster [USB-0] Setup...

Device: @1: 10CL120(Y[Z]/EP: Scan Chain

File: s/Quartus21_1/output_files/Test.sof ...

1. Export the content of the physical **DTCM** memory to a **DTCM.hex** file

2. Open the application in RARS and run it until reaching its endpoint.

3. Create a **DTCM.hex** file from RARS (at the application's endpoint).

4. Use TextDiff.exe application to validate the equality of the content of both files **DTCM.hex** from clauses 1 and 3.

Run a new application
(return on steps 1-4)